

Proofs

Theorem 1.1. (The Division Theorem for $\mathbb{Z}$): $a, b \in \mathbb{Z}$ with $b > 0 \exists! q, r \in \mathbb{Z}$ with $0 \leq r < b$ s.t. $a = bq + r$.

Proof. (Existence): Let $R = \{a - bm : m \in \mathbb{Z} \text{ and } a - bm \geq 0\}$, then as: if $a \geq 0$, $a - b \cdot 0 = a \geq 0$ so $a \in R$; if $a < 0$, $a - b \cdot a = a(1 - b) \geq 0$ so $a - b \cdot a \in R$, $R \neq \emptyset$. Thus, take $r = \min(R)$ & so $0 \leq r = a - bm \in R$ for some $m \in \mathbb{Z}$, choose $q = m \in \mathbb{Z}$. So by construction $a = bq + r$. Note, $r < b$ as were such not true then $r > a - b(q + 1) = r - b \in R$ but this contradicts $r = \min(R)$, so we have $0 \leq r < b$. (Uniqueness): Suppose $q, q', r, r' \in \mathbb{Z}$ with $0 \leq r', r < b$ & $a = bq + r = bq' + r'$, also WLOG take $r' \leq r$. So with $\frac{r-r'}{b} \in (-1, 1)$, & $\frac{r-r'}{b} = \frac{a-bq-(a-bq')}{b} = q' - q \in \mathbb{Z}$, we have $\frac{r-r'}{b} = 0$ i.e. $r = r'$. So $q = \frac{a-r}{b} = \frac{a-r'}{b} = q'$. $\square$

Euclidian Algorithm:

Let $a, b \in \mathbb{Z}^+$, apply Th^m 1.1, to give:

$$\begin{aligned} a &= bq_1 + r_1 \\ b &= r_1q_2 + r_2 \\ r_1 &= r_2q_3 + r_3 \\ &\vdots \end{aligned}$$

$$r_{n-1} = r_n q_{n+1} + 1,$$

Here, $\gcd(a, b) = r_n$. To prove this relation

$$G(r_i, r_{i+1}) = G(r_{i+1}, r_{i+2}),$$

where

$$G(a, b) = \{n \in \mathbb{Z} : n|a \text{ \& \& } n|b\}.$$

Reverse Euclid:

Given a completed algorithm we can obtain $n, m \in \mathbb{Z}$ s.t. $na + mb = \gcd(a, b)$ i.e. taking the end of a Euclidian algoithm:

$$\begin{aligned} &\vdots \\ r_{n-3} &= r_{n-2}q_{n-1} + r_{n-1} \quad (1) \\ r_{n-2} &= r_{n-1}q_n + r_n \quad (2) \\ r_{n-1} &= r_n q_{n+1} \\ \text{we have:} \\ (2) \Rightarrow r_n &= r_{n-2} - r_{n-1}q_n \quad (3) \\ (1) \Rightarrow r_{n-1} &= r_{n-3} - r_{n-2}q_{n-1} \quad (4) \\ (3) \& (4) \Rightarrow r_n &= r_{n-2} - (r_{n-3} - r_{n-2}q_{n-1})q_n \\ &\vdots \end{aligned}$$

we continue this ladder up, simplifying each time, (Note: $r_n = \gcd(a, b)$)

Theorem 1.9. Every integer $n \geq 2$ can be written as a product of primes.

Proof. (By strong induction):

(Base case, $n = 2$): Trivially true, $n = 2$ a product of primes.

(Inductive Hypothesis): Suppose for $n \in 2, 3, \dots, k$

$$n = p_{n,1}p_{n,2} \cdots p_{n,m_n}.$$

Then for $n = k + 1$, if $k + 1$ is prime we are done, if not then $k + 1 = ab$ where $a, b \in \{2, 3, \dots, k\}$, thus $a = p_{a,1}p_{a,2} \cdots p_{a,m_a}$ & $b = p_{b,1}p_{b,2} \cdots p_{b,m_b}$ so: $k + 1 = p_{a,1}p_{a,2} \cdots p_{a,m_a}p_{b,1}p_{b,2} \cdots p_{b,m_b}$. So $\forall n \in \mathbb{N}$ with $n \geq 2$, we are done. $\square$

Theorem 1.10. (Euclid's property of primes) $\forall p \in \text{Primes} \& \forall a, b \in \mathbb{Z}$. $p|ab \implies p|a \text{ or } p|b$.

Proof. (Case $p|a$): We are done. (Case $p \nmid a$): Then using the Euclidian algorithm we obtain: $1 = na - mp$ with $n, m \in \mathbb{Z}$. So $p|b \iff p|(na - mp) \iff p|n(ab) - mbp \iff p|ab$. $\square$

the chain of implications uses a proposition in the notes which states:

$$p|a \& p|b \implies p|a + b \& p|ab.$$

Theorem 1.12. (Fundamental Theorem of Arithmetic) $\forall n \in \mathbb{N}$ with $n \geq 2$, $\exists! p_1, p_2, \dots, p_r \in \text{Primes}$ up to ordering, s.t.: $n = p_1 p_2 \cdots p_r$. *Proof.* The existence is from Th^m 1.9. For uniqueness, suppose $p_1 p_2 \cdots p_r = q_1 q_2 \cdots q_s$, then cancel common primes, suppose the result is not 1 i.e. we are left with $p_1 p_2 \cdots p_n = q_1 q_2 \cdots q_m$ with $p_i \neq q_j, \forall i \in \{1, \dots, n\}$ & $\forall j \in \{1, \dots, m\}$, after reindexing. Then $p_1|q_1 q_2 \cdots q_m$ so $\exists j \in \{1, \dots, m\} p_1|q_j$, but q_j is prime so $p_1 = q_j, \hat{q}$. $\square$

Lemma 1.16. $\forall a, b \in R$:

$$\begin{aligned} 0a &= a0 = 0; \\ a(-b) &= (-a)b = -(ab); \& \\ (-a)(-b) &= ab \end{aligned}$$

Proof.

$$\begin{aligned} 0a + 0 &= 0a = (0 + 0)a = 0a + 0a \\ \text{so } 0a &= 0. \end{aligned}$$

$$\begin{aligned} ab + a(-b) &= a(b - b) = a0 = 0 = \\ (a - a)b &= ab + (-a)b \text{ so by the Groups \& } \text{Geo}^m \text{ we have } -(ab) \text{ is the unique element s.t.} \end{aligned}$$

$$\begin{aligned} ab - (ab) &= 0, \\ a(-b) &= -(ab) = (-a)b. \\ (-a)(-b) &= -(-a(-b)) = \\ &= -(-(ab)) = ab. \end{aligned} \quad \square$$

Lemma 1.18. (Subring Test) Let R be a ring & $S \subseteq R$. Then S is a subring of R iff the following hold, $\forall r, s \in S$:

1. $1_R \in S$,
2. $r + s, rs \in S$, &
3. $-r \in S$.

Proof. By 1 & 3 we have $-1_R \in S$ so $0_R = 1_R - 1_R \in S$ by 2. So $(S, +)$ is a group by the subgroup test with $(R, +)$. Commutativity of $+$, associativity of $+$ & $\times$, & distributivity of $\times$ over $+$ are all inherited from R . Note, as they shair operations they shair the identities of the operations, so as, $0_R \neq 1_R$ in R we have $0_S = 0_R \neq 1_R = 1_S$ in S , so $(R1)-(R4)$ are satisfied. $\square$

Lemma 2.2. Suppose $\text{Char}(R) = n > 0$, then: (i) $n \cdot r = 0, \forall r \in R$; & (ii) if $m \in \mathbb{Z}^+$ then $m \cdot 1 = 0$ iff $n|m$.

Proof. (i):

$$\begin{aligned} n \cdot r &= \underbrace{r + r + \cdots + r}_{n \text{ times}} \\ &= \underbrace{r(1 + 1 + \cdots + 1)}_{n \text{ times}} = r0 = 0 \end{aligned}$$

(ii): ($\implies$): Let $m \cdot 1 = 0$ with $m \in \mathbb{Z}^+$, applying the division Th^m with $m = nq + r$ where $q, r \in \mathbb{Z}$ & $0 \leq r < n$ so $0 = m \cdot 1 = (nq + r) \cdot 1 = q(n \cdot 1) + r \cdot 1 = q \cdot 0 + r \cdot 1 = r \cdot 1$, so $r = 0$ as otherwise $n > r \in \mathbb{Z}^+$ & $r \cdot 1 = 0$, a $\hat{z}$ of n 's defth. (*impliedby*): Let $n|m$ i.e. $\exists q \in \mathbb{Z}$ s.t. $m = qn$, so $m \cdot 1 = qn \cdot 1 = q(n \cdot 1) = q \cdot 0 = 0$. $\square$

Lemma 2.5. If S is a subring of R & R is a domain then S is a domain.

Proof. Let $r, s \in S$ with $rs = 0$, thus $rs = 0$ in R , hence $r = 0$ or $s = 0$. $\square$

Proposition 2.7. Suppose R is a domain. Then $R[X]$ is a domain.

Proof. We use the contrapositive of the definition of a domain [i.e.: R a domain iff $\forall 0 \neq r, s \in R, rs \neq 0$]. Let $0 \neq f, g \in R[X]$, taking the leading coefficients of f & g as $a \neq 0$ & $b \neq 0$, respectively, we the leading coefficient of fg as $ab \neq 0$, hence $fg \neq 0$. $\square$

Note: This immediately (by induction on n) yields $R[X_1, \dots, X_n]$ & domain if R is $\mathbb{D}_{\text{non-zero elements}}$.

Lemma 2.11. If R is a ring & $r \in R$ has both a left & right inverse then these equal & unique.

Proof. (Equal): Choose, $rs = 1 = tr$, then: $s = 1s = trs = t1 = t$. (Unique right inverse): Let $rs = 1 = rs'$ then $s = 1s = trs = t1 = trs' = 1s' = s'$, where t is choosen s.t. $tr = 1$. (Unique left inverse): Let $tr = 1 = t'r$ then $t = t1 = trs = 1s = t'rs = t'1 = t'$, where s is choosen s.t. $rs = 1$ $\square$

Proposition 2.13. Every division ring is a domain (& subsequently every field is an integral domain).

Proof. Let R be a division ring & suppose $r, s \in R$ & $rs = 0$. Then assume $r \neq 0$ so r^{-1} exists, then $s = r^{-1}rs = r^{-1}0 = 0$. Then given the immediate other case of $r = 0$ in all cases either $r = 0$ or $s = 0$. Hence R is a domain. $\square$

Lemma 2.12. The following conditions for an integer $n \geq 2$ are equivalent:

- (i) $\mathbb{Z}_n$ is an integral domain;
- (ii) $\mathbb{Z}_n$ is a field;
- (iii) n is prime.

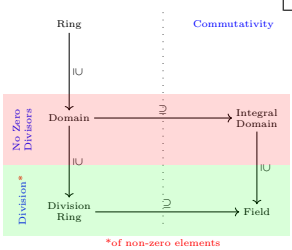
Proof. (i) $\implies$ (ii): Suppose $\mathbb{Z}_n$ is a domain & let $[m]_n \in \mathbb{Z}_n$ & $[m]_n \neq 0$. Consider the list $[0]_n[m]_n, [1]_n[m]_n, \dots, [n-1]_n[m]_n$, each element in the list is distinct as, suppose otherwise then $[i]_n[m]_n = [j]_n[m]_n$ for some distinct $i, j \in \{0, 1, \dots, n-1\}$, then $([i]_n - [j]_n)[m]_n = 0$ but $\mathbb{Z}_n$ is suppose to be a domain & $[m]_n \neq 0$ so $[i]_n - [j]_n = [0]_n$ i.e. $[i]_n = [j]_n, \hat{z}$. Then as there are n elements of the list each in $\mathbb{Z}_n$, then $\{[0m]_n, [1m]_n, \dots, [(n-1)m]_n\} = \mathbb{Z}_n$. So some multiple of $[m]_n$ say $[r]_n \in \mathbb{Z}_n$ is s.t. $[m]_n[r]_n = [r]_n[m]_n = [1]_n$. (ii) $\implies$ (iii): Suppose $\mathbb{Z}_n$ is a field then suppose $m, l \in \mathbb{N}$ are s.t. $ml = n$, so $0 < m \leq l \leq n$. Then $[m]_n[l]_n = [ml]_n = [n]_n = [0]_n$ so as fields are domains, $[m]_n = [0]_n = [n]_n$ or $[l]_n = [0]_n = [n]_n$ & so, by the equivalence class definition, $m = n$ or $l = n$ as neither are 0. So if $m = n, l = 1$ & if $l = n, m = 1$, so the only factors of n are 1 & n in $\mathbb{N}$, i.e. n is prime. (iii) $\implies$ (i): Suppose n is prime & let $[m]_n, [l]_n \in \mathbb{Z}_n$ s.t. $[m]_n[l]_n = 0$ i.e. $[ml]_n = [0]_n$ definitionally $n | ml$ so by Th^m 1.10 $n | m$ or $n | l$ i.e. $[m]_n = [0]_n$ or $[l]_n = [0]_n$. So with $\mathbb{Z}_n$'s commutativity, $\mathbb{Z}_n$ is an integral domain. $\square$

Proposition T1. In a domain R if $ab = ac$ or $ba = ca$ for $a, b, c \in R$ & $a \neq 0$ then $b = c$.

Proof. If $ab = ac$, then $a(b - c) = 0$. If $ba = ca$, then $(b - c)a = 0$. Since $a \neq 0$ and R is a domain, $b - c = 0$. $\square$

Lemma 2.15. In a ring R , the set of units R^* forms a group under multiplication.

Proof. (G0): $1 \cdot 1 = 1$ so $1 \in R^*$ i.e. $R^* \neq \emptyset$; (G1): given (R2) we have associativity & given $r, s \in R^*$, $(rs)(s^{-1}r^{-1}) = r(ss^{-1})r^{-1} = 1$ & $(s^{-1}r^{-1})(rs) = s^{-1}(r^{-1}r)s = 1$, so $rs \in R^*$ as required; (G2): (R4); & (G3): Let $r \in R^*$ so r is a unit so $r^{-1} \in R^*$ exists, which itself has inverse $(r^{-1})^{-1} = r \in R$, so $r^{-1} \in R^*$, as required. $\square$



Lemma 3.3. Suppose $\varphi : R \rightarrow S$ is an isomorphism. Then: (i) $\varphi(1_R) = 1_S$; (ii) $\varphi(0_R) = 0_S$; (iii) $\varphi(-r) = -\varphi(r), \forall r \in R$; (iv) $r \in R$ is invertible iff $\varphi(r) \in S$ is invertible and, in that case, $(\varphi(r))^{-1} = \varphi(r^{-1})$; (v) $r \in R$ is nilpotent iff $\varphi(r)$ is nilpotent and, in that case, they have the same index of nilpotence.

Proof. (i): Let $s \in S$. Since φ is bijective, $s = \varphi(r)$ for some $r \in R$. Then $\varphi(1_R)s = \varphi(1_R)\varphi(r) = \varphi(r) = s$ and similarly $s\varphi(1_R) = s$, hence $\varphi(1_R) = 1_S$. (ii): $\varphi(0_R) + 1_S = \varphi(0_R) + \varphi(1_R) = \varphi(1_R) = 1_S$, so $\varphi(0_R) = 0_S$. (iii): $\varphi(-r) + \varphi(r) = \varphi(0_R) = 0_S$, so $\varphi(-r) = -\varphi(r)$. (iv): If r is invertible then $\varphi(r^{-1})\varphi(r) = \varphi(r^{-1}r) = 1_S$ and $\varphi(r)\varphi(r^{-1}) = \varphi(rr^{-1}) = 1_S$, so $\varphi(r)$ is invertible with $(\varphi(r))^{-1} = \varphi(r^{-1})$. Conversely, apply this to $\varphi^{-1} : S \rightarrow R$. (v): Since $(\varphi(r))^n = \varphi(r^n)$, $r^n = 0_R$ iff $(\varphi(r))^n = 0_S$. Thus the least such n is the same. $\square$

Definitions

Definition 1.4. $\forall a, b \in \mathbb{Z}$, $\gcd(a, b) = \max\{n \in \mathbb{Z} : n|a \& n|b\}$, noting $\gcd(0, 0) = 0$

& if $\gcd(a, b) = 1$ then a & b are coprime.

Definition 1.6. $\forall a, b, n \in \mathbb{Z}$ with $n \geq 2$. a is **congruent to b modulo n** iff $n|a - b$, written $a \equiv b \pmod n$. Then define the congruence class of a w.r.t. n as: $[a]_n = \{b \in \mathbb{Z} : b \equiv a \pmod n\} = \{b \in \mathbb{Z} : n | (b - a)\} = \{a + kn : k \in \mathbb{Z}\}$. We thus have:

$\mathbb{Z}_n = \{[a]_n : a \in \mathbb{Z}\} = \{[0]_n, [1]_n, \dots, [n-1]_n\}$
Note: $b \equiv a \pmod n$ iff $b = a + kn$ iff $[a]_n = \{a + nk : k \in \mathbb{Z}\}$ iff $[a]_n = [a + kn]_n$

Definition 1.14. A **ring** is a set R with two binary operations $+$ & $\times$, defined on R , satisfying the following:

- (R1) $(R, +)$ is an abelian group with identity 0.
- (R2) $\times$ is associative.
- (R3) $\times$ distributes over $+$ i.e.: $a(b + c) = ab + ac$, $(a + b)c = ac + bc, \forall a, b, c \in R$
- (R4) The multiplicative identity $1 \in R$ & $1 \neq 0$

Example. With $n \geq 2$, $M_n(R)$ is the ring of $n \times n$ matrices, a non-commutative ring. Note: the only two-sided ideals of $M_n(F)$, F a field, are $M_n(F)$ & $\{0\}$ but this isn't a division ring hence even relaxing commutative in Prop³ 3.22 doesn't yeild the same prop³ with field $\rightarrow$ division ring.

Example. For X a set $\mathcal{P}(X)$ with $A + B = (A \cup B) \setminus (A \cap B)$, $AB = A \cap B$

& $0 = \emptyset, -A = A, A, B \in \mathcal{P}(X)$.

Definition 1.17. Let R be a ring & $S \subseteq R$. Then S is a **subring** of R if S is a ring w.r.t. the same addition & multiplication as R & contains the multiplicative identity of R , 1_R .

Definition 1.21. let R be a ring. The **ring of polynomials**

$R[X] = \{\sum_{i \geq 0} a_i X^i \text{ a finite sum} : a_i \in R\}$ with addition & multiplication defined as: $\sum_{i \geq 0} a_i X^i + \sum_{i \geq 0} b_i X^i = \sum_{i \geq 0}^{\max(n, m)} (a_i + b_i) X^i$ & $(\sum_{i \geq 0} a_i X^i)(\sum_{i \geq 0} b_i X^i) = \sum_{i \geq 0}^{n+m} (\sum_{j=0}^i a_j b_{i-j}) X^i$

$0 = \sum_{i \geq 0} 0X^i$ & $1 = 1X^0 + \sum_{i \geq 1} 0X^i$. Then the

degree of a polynomial is the greatest $i \in \mathbb{N} \cup \{0\}$ s.t. $a_i \neq 0$ or if no such i exist $\deg(0) = -\infty$

More genearily, we have:

$$R[X_1, \dots, X_n] = R[X_1, \dots, X_{n-1}][X_n]$$

$$\forall f, g \in R[X], \deg(f + g) \leq \max(\deg(f), \deg(g)) \& \deg(fg) \leq \deg(f) + \deg(g).$$

Definition 1.24. If R_1 & R_2 are rings then the **Cartesian product** $R_1 \times R_2$ with the $+$ & $\times$ defined element wise is a ring.

Definition 2.1. For $a \in R$ define: for $n \in \mathbb{Z}^+$, $n \cdot a = \underbrace{a + a + \cdots + a}_{n \text{ times}}$; for $n = 0$,

$$\begin{aligned} n \cdot a &= 0 \cdot a = 0; \& \text{ for } n \in \mathbb{Z}^-, \\ n \cdot a &= (-n) \cdot (-a). \end{aligned}$$

The **Characteristic**, $\text{char}(R)$, of a ring R is the least positive integer n s.t. $n \cdot 1 = 0$, or if no exist $\text{char}(R) = 0$.

Note: $(n + m) \cdot a = n \cdot a + m \cdot a$, $n \cdot (a + b) = n \cdot a + n \cdot b$, & $n \cdot (m \cdot a) = (nm) \cdot a$, all easily verifiable.

Definition 2.3. A non-zero element $r \in R$ is a **zero-divisor** if $\exists s \in R$ a non-zero element s.t. $rs = 0$ or $sr = 0$.

A ring R is a domain if, $\forall r, s \in R$:

$$rs = 0 \implies r = 0 \text{ or } s = 0.$$

i.e. a ring with no zero divisors is a domain & vice versa.

Definition 2.9. A **division ring** is a ring where every non-zero element has a right & left inverse, i.e.: $\forall r \in R \setminus \{0\}, \exists s, t \in R$ s.t. $rs = 1 = tr$ with s, t being the right & left inverses, respectively.

Note: this immediately collapse, due to Lem^m 2.11, to just there exists an inverse as $s = t$.

An **invertible** element r is called a **unit**.

Example. Division rings are not all fields e.g. the quaternions $\mathbb{H} = \{a + bi + cj + dk : a, b, c, d \in \mathbb{R}\}$ where $i^2 = j^2 = k^2 = -1$ &

$ij = k, jk = i, \& ki = j$, this is non-commutative as

$$-ij = -k = jk = ji \text{ so } ij \neq ji$$

and this is a division ring as for $z = a + bi + cj + dk$

$$z^{-1} = \frac{a-bi-cj-dk}{a^2+b^2+c^2+d^2}, \text{ taking } z \neq 0.$$

Definition 2.16. An element r of a ring R is **nilpotent** if $\exists n \in \mathbb{Z}^+$ s.t. $r^n = 0$, and the least such n is the **index of nilpotence** of r . An element $r \in R$ is **idempotent** if $r^2 = r$.

Note: In any ring R , 0_R is nilpotent with index 1 and 0_R and 1_R are idempotent.

Definition 3.1. If R & S are rings then an **isomorphism** from R to S is a bijection $\varphi : R \rightarrow S$ s.t. $\forall r, r' \in R$:</